

AVERAGE REWARD

TD error:

$$\delta_t = R_{t+1} - \bar{R}_{t+1} + \hat{V}_\pi(S_{t+1}, \mathbf{w}_t) - \hat{V}_\pi(S_t, \mathbf{w}_t)$$

For action-values and SARSA:

$$\delta_t = R_{t+1} - \bar{R}_{t+1} + \hat{q}_\pi(S_{t+1}, A_{t+1}, \mathbf{w}_t) - \hat{q}_\pi(S_t, A_t, \mathbf{w}_t)$$

Semi-gradient SARSA update:

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \delta_t \nabla \hat{q}_\pi(S_t, A_t, \mathbf{w}_t)$$

Differential Semi-Gradient SARSA

Initialize S, A ; $\bar{R} = 0$.

At each time step:

Execute action A . Observe S', R .

Choose $A' \sim \epsilon$ -greedy over values $\hat{q}_\pi(S', a, \mathbf{w})$, $\forall a$.

$$\delta = R - \bar{R} + \hat{q}_\pi(S', A', \mathbf{w}) - \hat{q}_\pi(S, A, \mathbf{w})$$

$$\bar{R} = \bar{R} + \beta \delta \quad \left[\begin{array}{l} \text{This makes sense as} \\ \delta \text{ is a sample estimate} \\ \text{of average reward } r(\pi). \\ \text{(Show R2 above).} \end{array} \right]$$

$$\mathbf{w} = \mathbf{w} + \alpha \delta \nabla \hat{q}_\pi(S, A, \mathbf{w})$$

$$S = S', A = A'$$

$$\begin{aligned} \bar{R} &= \bar{R} + \beta \delta \\ &= \bar{R} + \beta (R - \bar{R} + \hat{q}_\pi(S', A', \mathbf{w}) - \hat{q}_\pi(S, A, \mathbf{w})) \\ &= (1 - \beta) \bar{R} + \beta (R + \hat{q}_\pi(S', A', \mathbf{w}) - \hat{q}_\pi(S, A, \mathbf{w})) \end{aligned}$$

Consider

$$q(S, a) = \mathbb{E}_\pi [R_t - r(\pi) + q(S', a') \mid S_t = S, A_t = a]$$

$$\Rightarrow q(S, a) = \mathbb{E}_\pi [R_t + q(S', a')] - r(\pi)$$

$$\Rightarrow r(\pi) = \mathbb{E}_\pi [R_t + q(S', a') - q(S, a)]$$

$\Rightarrow R_t + q(S', a') - q(S, a)$ is
a sample estimate of $r(\pi)$.